

CAEM: Confidence-Aware Episodic Memory with Self-Improvement

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Abstract

Large language models produce confident but factually incorrect outputs at rates that span three to ninety-one percent across deployment domains, a failure mode that increased parameter count alone does not close. This thesis proposes Confidence-Aware Episodic Memory with Self-Improvement (CAEM), an architectural intervention that treats hallucination reduction as a system-level problem rather than a post-hoc detection problem. A single query passes through a verified episodic memory store, a confidence-aware three-tier router with an independent safety override, and a nine-signal verifier ensemble. Against the matched zero-shot baseline, the within-trajectory best cycle reaches a pooled exact-match gain of +20.3% and a pooled composite hallucination metric reduction of −29.4% on the training panel. Against the locked external eight-baseline panel, CAEM leads on both pooled axes at the best and final cycles.

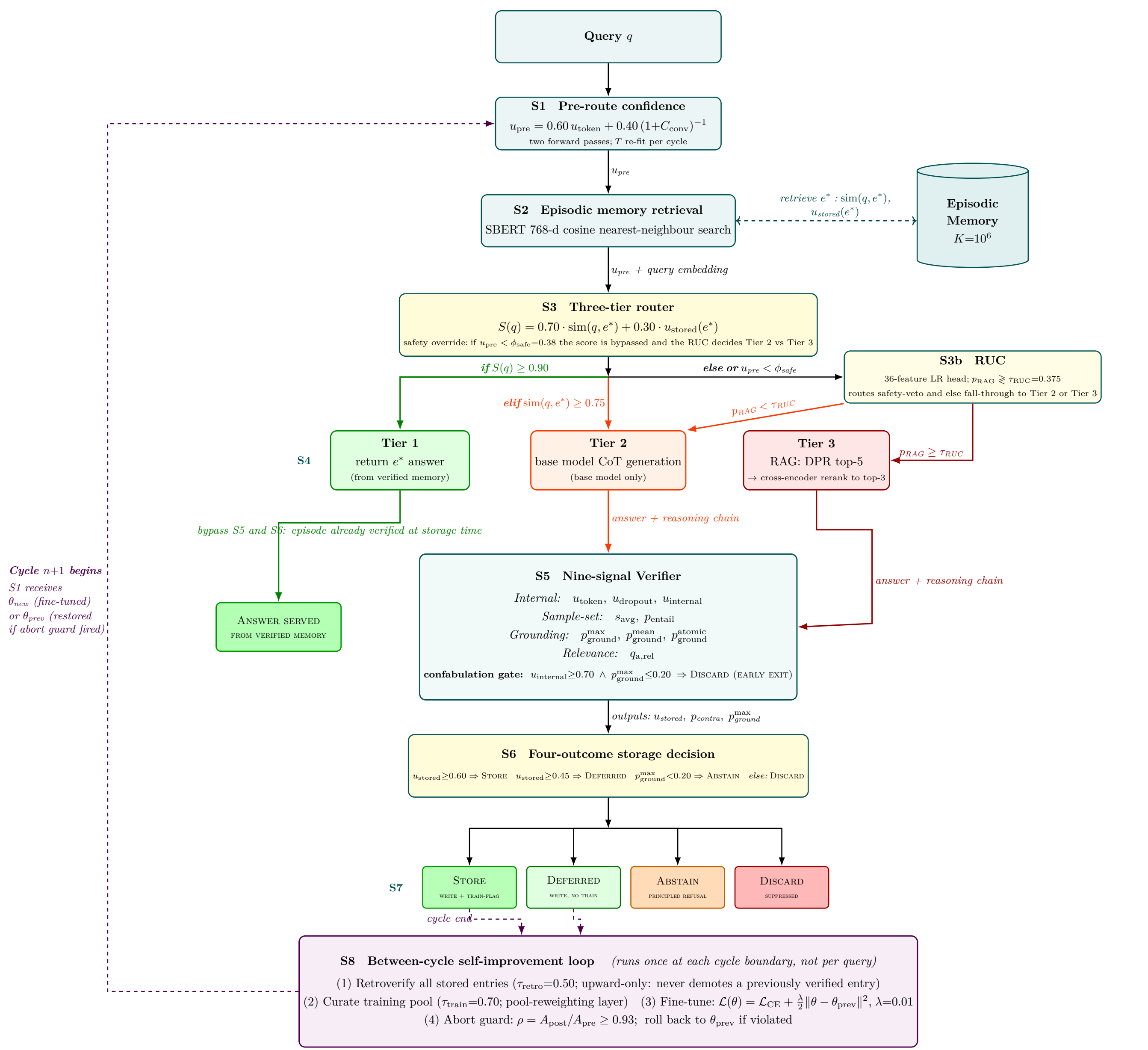
Research Objective

- Verified episodic memory with a four-outcome decision:** design an episodic memory whose admission rule is finer than a single binary threshold.
- Pre-generation confidence, three-tier routing with safety override, and a retrieval-utility classifier:** compute a pre-generation confidence estimate from a single short forward pass on the query that fuses a short-greedy-decode token-probability aggregate with an encoder-layer convergence ratio, calibrate it via per-benchmark temperature scaling on a disjoint fold, and use it independently from the memory similarity score and the stored-answer quality.
- Nine-signal post-generation verifier with a calibrated probability composite:** combine nine complementary signals into a per-answer record across four families: internal calibration of the generator, sample-set agreement across resampled chains, external grounding against retrieved evidence under three aggregation rules, and question-answer relevance.
- Fixed-threshold storage gate with two registered cut-points:** convert the calibrated probability composite into the storage and deferral decisions through a fixed-threshold rule whose two cut-points and one abstention floor are registered before the main run begins and held constant across cycles.
- Constrained self-improvement with a parameter-efficient adapter and a multi-modal retention guard:** curate the fine-tuning pool from stored episodes that exceed a strict quality threshold set above the storage cut-point and pass the curated pool through a benchmark-balancing reweighting layer that prevents single-benchmark dominance.

Methodology

The architecture organises a single query into an eight-stage pipeline traversed once per query and wrapped in a per-cycle update loop. Stage 1 produces a pre-routing confidence from a single short forward pass. Stage 2 encodes the query and probes a similarity-indexed episodic memory for the nearest stored entry. Stage 3 combines the similarity and stored-quality scores into a dispatch score and routes to one of three tiers, with a safety override on low confidence. Stage 3b, a retrieval-utility classifier, refines the dispatch into zero-shot or retrieval-augmented tier per query. Stage 4 executes the chosen tier; Stage 5 passes the answer through a nine-signal post-generation verifier; Stage 6 applies a four-outcome decision (store, defer, abstain, discard); Stage 7 returns the user-facing action. Stage 8, at cycle boundaries, runs retroactive re-verification, deferred reconsideration, memory consolidation, and a regularised fine-tune under a retention guard with asymmetric rollback: weights are rolled back on retention failure, but memory is preserved.

Proposed Model Architecture



Cycle $n+1$ begins
S1 receives
 θ_{new} (fine-tuned)
or θ_{prev} (restored
if short guard fired)

bypass S5 and S6: episode already verified at storage time

cycle end

The eight-stage pipeline routes each query through pre-routing confidence, memory lookup, three-tier dispatch with the Stage 3b retrieval-utility classifier, tier execution, nine-signal verification, the four-outcome decision, and output; Stage 8 closes every cycle boundary with retroverify, deferred reconsider, consolidate, LoRA fine-tune, and the multi-modal retention probe.

Locked Configuration

Component	Value
Storage cut τ_{store} / τ_{defer}	0.60 / 0.45
SIL admission τ_{train}	0.70
Retrieval-utility classifier τ_{RUC}	0.375
Retention floor ρ_{min}	0.93
Router λ / safety ϕ_{pg}	0.70 / 0.38
Backbone	Qwen-2.5-3B-Instruct (frozen)
LoRA	$r=32$, $\alpha=64$, 7 projections
Active verifier signals	9
Trajectory	registered 10 / realised 6 cycles

Per-Cycle Trajectory

Cycle	Pooled EM	Pooled CHM	MMLU	TQA-test	CSQA-test	Outcome
B1 (zero-shot ref)	0.507	0.120	—	—	—	reference
C0	0.541	0.135	—	—	—	baseline

Headline Result

Against the matched zero-shot baseline B1 on the same backbone, the training-panel architectural-contribution slice at the within-trajectory best cycle (C2) reaches a pooled exact-match gain of +20.3% relative and a pooled composite hallucination metric reduction of −29.4% relative; the trajectory-end cycle holds at +13.7% EM and −20.3% CHM. CAEM beats every external baseline on both axes at both reference cycles.

$$\Delta EM = +20.3\% \text{ rel.}$$

$$\Delta CHM = -29.4\% \text{ rel.}$$

(training panel, CAEM C2 vs. B1 zero-shot baseline)

External Baseline Comparison

System	EM	CHM	ΔEM	ΔCHM
CAEM C2 *	0.544	0.107	—	—
CAEM C5 (final)	0.517	0.109	−0.027	+0.002
B1 zero-shot	0.507	0.120	+7.4%	−11.0%
B2 chain-of-thought	0.500	0.123	+8.8%	−12.7%
B3 DPR-RAG	0.458	0.148	+18.8%	−27.7%
B4 CoT+DPR-RAG	0.461	0.142	+17.9%	−24.6%

System	EM	CHM	ΔEM	ΔCHM
B5 five-shot CoT	0.405	0.136	+34.2%	−21.4%
B6 vanilla FT (C5)	0.325	0.144	+67.2%	−25.5%
B7 FLARE	0.292	0.153	+86.3%	−29.9%
B8 semantic entropy	0.505	0.120	+7.7%	−10.5%

Wins-against (joint ΔEM & ΔCHM):
CAEM C2 beats 8/8; CAEM C5 beats 8/8.

Retrieval-Utility Classifier Ablation

Benchmark	EM off	EM on	ΔEM (pp)	T3 off	T3 on
FEVER	0.503	0.503	0.0	92.3%	70.7%
TriviaQA	0.437	0.463	+2.7	95.7%	68.0%
CommonsenseQA	0.603	0.743	+14.0	57.7%	20.3%
StrategyQA	0.617	0.640	+2.3	100.0%	76.7%
TruthfulQA	0.210	0.310	+10.0	99.0%	40.0%
Pooled	0.474	0.532	+5.8	88.9%	55.1%

References

- [1] Guo et al. *ICML 2017* (calibration). [2] Lin et al. *ACL 2022* (TruthfulQA). [3] Thorne et al. *NAACL 2018* (FEVER). [4] Joshi et al. *ACL 2017* (TriviaQA). [5] Talmor et al. *NAACL 2019* (CommonsenseQA). [6] Hu et al. *ICLR 2022* (LoRA). [7] Tang et al. *ACL 2024* (MiniCheck). [8] Cherian et al. *NeurIPS 2024* (boost layer).